Biologically informed blood-based risk stratification for chronic pain in insomnia

Keywords: Hematological biomarkers | Inflammation | Hypothalamic–pituitary–adrenal axis | Machine learning | Risk stratification

Abstract

Objective: To develop and validate a blood-based model for identifying current chronic pain status among patients with insomnia using routinely available laboratory markers.
Methods: We analyzed 481 clinically diagnosed insomnia patients (143 with chronic pain). Six biomarkers (IL-6, IL-10, TNF-α, CRP, cortisol, ACTH) were used to construct a biologically informed 15-feature hematological panel by adding derived ratio features. Models were developed on the training set and evaluated once on an independent test set. Discrimination, probability estimation, and clinical utility were assessed using ROC/PR metrics, Brier score, calibration, and decision curve analysis (DCA), with ENS-RankPlatt (IL-10) and a six-biomarker blood model as reference comparators.
Results: On the test set, the final ensemble (HemoPain-Ensemble) achieved ROC-AUC 0.819, AP 0.826, and Brier 0.295, with favorable calibration and higher net benefit than treat-all/treat-none and both reference models over clinically relevant thresholds.
Conclusion: A biologically guided 15-feature blood panel combined with ensemble learning provides robust and clinically informative risk stratification for chronic pain in insomnia, with post-hoc interpretability and supplementary analyses supporting the contribution of ratio-based features.

1 | Introduction

Insomnia represents a significant and growing public health challenge worldwide. Recent global estimates indicate that about 16.2% (852.3 million) of adults aged ⩾20 years experienced chronic, clinically relevant insomnia, and nearly half of these individuals (414.9 million) had severe insomnia [1]. Chronic pain is one of the most common comorbidities in patients with insomnia, with a prevalence of around 50%, which is substantially higher than that in the general population (18%–30%) [2-5]. A growing body of evidence suggests a bidirectional relationship between insomnia and chronic pain, with a stronger impact of insomnia on chronic pain than the reverse [6-10]. Compared to insomnia alone, the coexistence of insomnia and chronic pain can exacerbate symptoms and complicates treatment strategies.

For example, individuals with comorbid insomnia and chronic pain demonstrate a higher likelihood of experiencing mental health disorders, such as depression and anxiety [11, 12]. Therefore, it is of substantial clinical relevance to identify objective blood-based markers that can assist risk stratification and classification of current chronic pain status in patients with insomnia.

The current diagnostic approach for chronic pain in patients with insomnia relies primarily on patients' subjective descriptions, professional scale assessments, and doctors' subjective judgements, with a lack of objective diagnostic markers. In clinical practice, however, this diagnostic process can be challenging. An important reason is that insomnia is frequently comorbid with mood disorders and cognitive alterations, which can influence the patients' symptom descriptions. For example, some patients may overemphasize insomnia while unintentionally downplaying or concealing pain issues. These potential factors can result in underdiagnosis or misdiagnosis of chronic pain, leading to delayed or inadequate treatment and prolonged suffering. Therefore, it is of great clinical significance to find objective blood-based markers for classification and risk stratification of current chronic pain status

To date, however, reports on the objective markers for identifying chronic pain in patients with insomnia are very limited. Screening diagnostic biomarkers from peripheral blood based on the underlying mechanisms of disease represents a core pathway in translational medicine research. Low-grade inflammation and hypothalamus-pituitary-adrenal (HPA) axis dysfunction are considered potential mechanisms linking insomnia to chronic pain [13]. Nevertheless, our current understanding of the pathogenesis rests on two pillars: experimental sleep deprivation studies, and the integration of findings from independent investigations into insomnia and chronic pain. Consequently, it remains unknown whether peripheral inflammatory factors and HPA axis hormones can serve as diagnostic biomarkers for chronic pain in patients with insomnia.

To address this gap, this study first evaluated the diagnostic value of the inflammatory and HPA axis biomarkers in peripheral blood—namely interleukin-6 (IL-6), interleukin-10 (IL-10), C-reactive protein (CRP), cortisol, and adrenocorticotropic hormone (ACTH)—for chronic pain in patients with insomnia. However, a single biomarker often plays a small role in disease pathology and fails to capture the complexity of molecular interactions, thereby limiting its diagnostic accuracy and increasing the risk of misclassification [14-16]. For this reason, composite biomarker panels may provide superior diagnostic performance. In recent years, machine learning has been widely used for disease diagnosis and biomarker discovery. Compared with traditional regression models, machine learning is more capable of capturing nonlinear relationships and complex interactions in biological systems without pre-specifying interaction terms, which is particularly important for studying pathophysiological process

involving multiple biological pathways [17-19]. Therefore, this study also aimed to develop and validate a machine-learning framework that integrates the aforementioned biomarkers to improve blood-based risk stratification of current chronic pain status in patients with insomnia. Additionally, to further improve prediction performance, we introduced a novel strategy of integrating biomarker ratios as new biomarkers into the machine learning framework since prior studies suggested that biomarker ratios may demonstrate superior diagnostic power compared to individual markers by capturing complex biological information related to biomarker balance, inter-pathway interactions, and physiological states [16, 20, 21].

In summary, this study evaluated the value of peripheral inflammatory markers and HPA-axis hormones for identifying current chronic pain status in patients with insomnia, and subsequently developed and validated machine-learning-based multi-biomarker models for blood-based risk stratification. These findings may deepen our understanding of the biological processes linking insomnia and chronic pain, and the proposed models may provide clinically useful support for classification and risk stratification of current chronic pain status in patients with insomnia.

2 | Methods

The overall study workflow is presented in Figure 1.

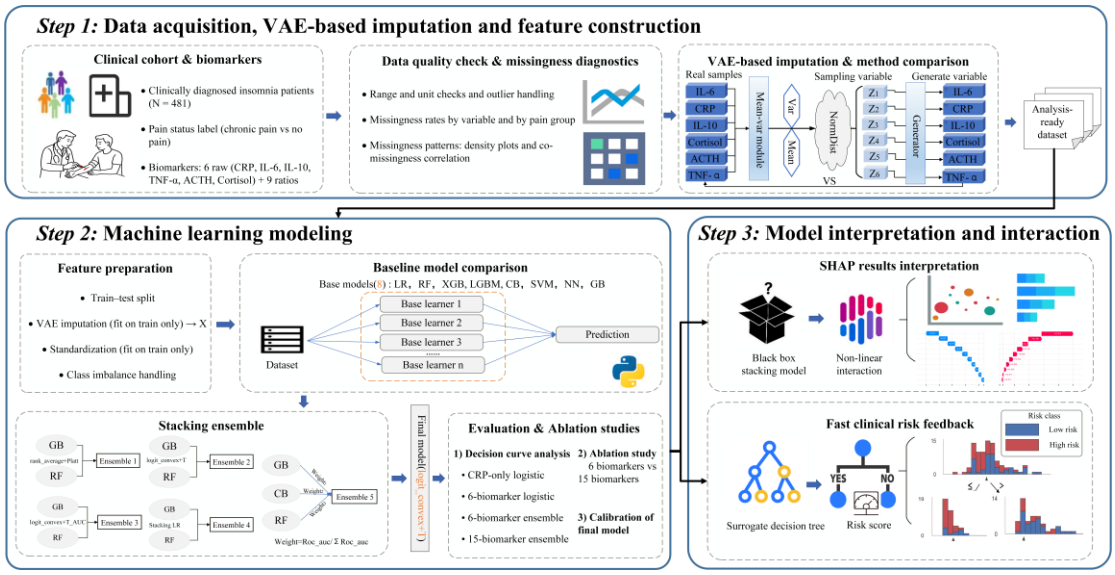

FIGURE 1 | Study workflow and model pipeline.

2.1 | Study population and data splitting

This study conducts a retrospective biological risk stratification analysis in patients with

clinically diagnosed insomnia [1]. Hematological profiles and standardized clinical assessments obtained at the baseline visit are included in the analysis [2]. Given the cross-sectional nature of the dataset, the objective is to identify blood-based markers associated with current chronic pain vulnerability [3].

To ensure temporal and analytical consistency, only candidate predictors available at the time of assessment are considered [4,5], whereas records with missing predictor values are retained to preserve the integrity of the original cohort [6]. Missing predictor data are handled during preprocessing, with imputation models fitted exclusively on the training set and then applied to the test set to avoid information leakage [7].

The dataset is randomly divided into a training set and an independent test set in an 8:2 ratio using stratified sampling [8]. A fixed random seed (seed = 42) is used to ensure reproducibility. The training set serves as the sole basis for all model development procedures, including hyperparameter tuning, model comparison, calibration, and threshold selection. The test set is reserved exclusively for out-of-sample evaluation and remains untouched throughout the model development process.

2.2 | Candidate predictors and feature engineering

Specifically, blood biomarkers are selected a priori based on clinical relevance, biological interpretability, and routine availability [9–11]. To ensure translational relevance, the predictor set is limited to hematological variables measured at the baseline visit [12].

Six primary laboratory indicators are included: C-reactive protein (CRP), interleukin-6 (IL-6), interleukin-10 (IL-10), tumor necrosis factor alpha (TNF-α), adrenocorticotropic hormone (ACTH), and plasma total cortisol (cortisol). Collectively, these markers capture key aspects of systemic inflammatory activity and hypothalamic–pituitary–adrenal (HPA) axis function [10,13–15].

Beyond absolute concentration, we pre-specify ratio-based features with input from clinical experts to reflect relative biological balance rather than isolated marker levels [11,16]. These ratios represent the quotient of biomarkers with physiological relevance to inflammatory signaling and neuroendocrine regulation [14–16].

Formally, each ratio-based feature is defined as the quotient of two pre-specified hematological biomarkers:

$$R_{i,j} = \frac{X_i}{X_j}, \quad (1)$$

where X_i and X_j denote the measured concentrations of two selected biomarkers reflecting inflammatory activity or hypothalamic–pituitary–adrenal (HPA) axis function, and $R_{i,j}$ represents the corresponding ratio feature.

This feature set is specified independent of outcome information and is implemented identically across both training and test sets [16].

In total, nine ratio features are derived from the primary biomarkers, yielding a final panel of 15 hematological features. The complete feature panel, including variable definitions and missingness rates, is detailed in Table 1.

TABLE 1 | Summary of primary indicators and derived ratio features with missingness diagnostics (N=481). Non-hematological variables, including demographic characteristics and clinical assessment scores, were not incorporated into the primary model. Their variable definitions and missingness rates are reported in the Supplementary Material (Supplementary Table S1) for completeness.

Category	Variable	Missing n(%)	Definition
Primary indicator	CRP	11(2.3%)	C-reactive protein(mg/L)
Primary indicator	Cortisol	31(6.4%)	Plasma total cortisol
Primary indicator	IL-6	10(2.1%)	Interleukin-6(pg./mL)
Primary indicator	IL-10	23(4.8%)	Interleukin-10(pg./mL)
Primary indicator	TNF-α	24(5.0%)	Tumor necrosis factor-α (pg./mL)
Primary indicator	ACTH	12(2.5%)	Adrenocorticotropic hormone (pg./mL)
Derived ratio	IL-6/IL-10	98 (20.3%)	Pro- vs anti-inflammatory balance
Derived ratio	TNF-α/IL-10	96 (19.9%)	Pro-inflammatory dominance
Derived ratio	CRP/IL-10	98 (20.3%)	Systemic vs anti-inflammatory balance
Derived ratio	Cortisol/ACTH	98 (20.3%)	HPA-axis adrenal responsiveness

Derived ratio	Cortisol/IL-6	116 (24.1%)	Neuroendocrine–inflammatory interaction
Derived ratio	Cortisol/CRP	99 (20.5%)	HPA-axis vs systemic inflammation
Derived ratio	IL-6/TNF-α	99 (20.5%)	Cytokine network balance
Derived ratio	CRP/IL-6	22 (4.6%)	Acute-phase vs cytokine status
Derived ratio	ACTH/IL-6	22 (4.6%)	Neuroendocrine–immune feedback

2.3 | Missing data handling and preprocessing

Ratio features were generated prior to imputation and treated as pre-specified derived predictors within the same preprocessing pipeline [17]. Missing values in both primary biomarkers and derived ratio features are then handled using training-set-fitted imputation models before downstream model development.

Prior to imputation, missingness is systematically characterized across all pre-specified hematological features [18,19]. For each variable X_j, the missing rate is quantified as

$$r_j = \frac{n_j^{miss}}{n}, \tag{2}$$

where n_j^{miss} denotes the number of missing observations for feature j, and n is the total sample size. Missingness patterns are examined at both the individual feature level and across feature combinations to assess potential structure in co-missingness. In addition, missing rates are compared between individuals with and without chronic pain to evaluate potential outcome-related bias. We report baseline feature distributions and missingness diagnostics prior to imputation in Supplementary Fig. S1.

Given the presence of substantial missingness, multiple imputation strategies are evaluated, including mean imputation, K-nearest neighbors (KNN), multivariate imputation by chained equations (MICE), and a mask-aware variational autoencoder (VAE) [17] [20–22].These methods were compared to balance imputation accuracy with preservation of biologically meaningful relationships among features [18].

For the VAE-based approach, the reconstruction loss is computed only on observed entries, using a binary mask matrix M [21], such that

$$L_{recon} = \sum_{i=1}^{n} \sum_{j=1}^{p} M_{ij} \ell(X_{ij}, \widehat{X}_{ij}), \tag{3}$$

where X_{ij} and $\widehat{X}_{ij}$ denote the observed and reconstructed values of feature j for individual i, respectively, and $\ell(\cdot)$ represents the reconstruction loss. All imputation models were trained exclusively on the training set, and the learned parameters were applied unchanged to the test set to avoid information leakage [22].

Imputation quality is evaluated using distributional fidelity and preservation of inter-feature relationships [17,22]. For each feature, similarity between observed and imputed distributions was assessed using the Kolmogorov–Smirnov statistic and the first-order Wasserstein distance,

$$D_{KS} = \sup_{x} \left| F_{obs}(x) - F_{imp}(x) \right|, \tag{4}$$

$$W_1 = \int_0^1 \left| F_{obs}^{-1}(u) - F_{imp}^{-1}(u) \right| du, \tag{5}$$

where F_{obs} and F_{imp} denote the empirical cumulative distribution functions of the observed and imputed data, respectively. To quantify systematic level shifts, the relative mean deviation is additionally computed

$$\Delta\mu = \frac{\mu_{imp} - \mu_{obs}}{\mu_{obs}}. \tag{6}$$

To assess whether multivariate dependency structures are preserved after imputation, Spearman correlation matrices are computed for both observed and imputed datasets. Differences in correlation structure were summarized using the average absolute deviation of pairwise correlations across feature pairs,

$$\Delta\rho = \frac{2}{p(p-1)} \sum_{j<k} \left| \rho_{jk}^{obs} - \rho_{jk}^{imp} \right|, \tag{7}$$

where ρ_{jk}^{obs} and ρ_{jk}^{imp} denote the Spearman correlation coefficients between features j and k before and after imputation, respectively.

Quantitative metrics and diagnostic visualizations are jointly considered to compare imputation strategies. For overall comparison across imputation methods, these feature-level error measures were standardized and aggregated into a composite imputation error score, with lower values indicating better overall fidelity.

Following imputation, continuous features are z-score standardized using training-set parameters and then applied unchanged to the test set [20,22].

2.4 | Model development and internal validation

Based on the finalized hematological feature set, machine learning models are developed to predict chronic pain status.

Eight candidate base learners are considered: logistic regression, support vector machines, random forest, gradient boosting, XGBoost, LightGBM, CatBoost, and multilayer perceptron neural networks. These models are chosen to represent commonly used linear and non-linear learners in clinical prediction tasks [23].

All models are trained and tuned using five-fold stratified cross-validation within the training set [24]. Hyperparameter optimization and internal model comparison are performed exclusively within this cross-validation framework. Model performance during development is evaluated using multiple complementary metrics, including the area under the receiver operating characteristic curve (AUC), the area under the precision–recall curve (AP), the Brier score, and the F1 score [25]. No single metric is used as the sole criterion for model selection. We report average precision (AP), defined as the area under the precision–recall curve, to complement ROC-AUC. AP is threshold-free and more informative than accuracy in moderately imbalanced settings [25].

After cross-validation, two base learners are selected according to a pre-specified training-set criterion that jointly considers discrimination, calibration-related metrics, and stability across folds [25,26]. The selected base learners are fixed prior to ensemble construction to define a constrained ensemble design.

Using the fixed base learner pair, five primary ensemble strategies are implemented: (i) logit-scale convex combination with temperature scaling optimized for log loss (ENS-LogitT), (ii) logit-scale convex combination with temperature scaling optimized for ROC-AUC (ENS-LogitT-AUC), (iii) rank-based averaging with Platt calibration (ENS-RankPlatt), (iv) probability-level averaging with post hoc calibration, and (v) logistic-regression-based stacking (ENS-Stack) [23,27]. For both ENS-LogitT and ENS-LogitT-AUC, ensemble predictions are formed by convex combination in the logit space followed by temperature scaling, using the fixed base learner pair of gradient boosting and random forest. The two methods differ in their optimization objectives: ENS-LogitT selects the ensemble weight and temperature by minimizing out-of-fold log loss, whereas ENS-LogitT-AUC selects these parameters by maximizing out-of-fold ROC-AUC. All ensemble parameters are determined using training data only [24,27]. In addition, a reference ensemble combining the top three individual models (ENS-Top3) is included for comparison.

Model selection and ensemble specification are finalized using training data only [26,27]. Specifically, candidate base learners and ensemble strategies are compared within the five-fold cross-validation framework in the training set, and the final modeling approach (hereafter referred to as HemoPain-Ensemble) is pre-specified based on cross-validated performance and stability across metrics. The independent test set is held out as a strict analytical boundary and is used only once for final out-of-sample evaluation [24]. For completeness, test-set performance of alternative candidates is reported descriptively, but the test set is not used for model selection, hyperparameter tuning, calibration, or threshold determination.

2.5 | Performance assessment and clinical utility

Performance assessment centers on the pre-specified final model, HemoPain-Ensemble, while comparative results for alternative candidates are reported separately [28,29].

To quantify predictive uncertainty, we apply non-parametric bootstrapping with 1,000 resamples to the test set [29,30]. For each resample, we repeat the evaluation procedure and record discrimination, calibration, and threshold-dependent metrics. We then derive point estimates and 95% confidence intervals from the bootstrap distributions [30]. No model refitting, recalibration, or re-selection of thresholds is performed within bootstrap resamples; the evaluation reuses the fixed model and fixed thresholds determined during training [29,31].

$$CI_{95\%} = \left[Q_{0.025}(\hat{\theta}^*),\ Q_{0.975}(\hat{\theta}^*) \right], \tag{8}$$

where $\hat{\theta}^*$ denotes the bootstrap estimate of a given performance metric and $Q_q(\cdot)$ represents the empirical q-th quantile.

Risk stratification uses decision thresholds that are pre-specified on the training set and then applied unchanged to the independent test set.[25,29] The primary threshold is selected to maximize the F1 score within five-fold cross-validation on the training set.[25] In addition, two alternative thresholds are defined on the training set to reflect different operating preferences: (i) a sensitivity-constrained threshold (Sen $\geqslant$ 0.80) and (ii) the Youden index – optimal threshold. All threshold-dependent metrics on the test set and across bootstrap resamples reuse these fixed thresholds.[27,29]

We assess clinical utility using decision curve analysis.[28] Net benefit is calculated as follows:

$$\text{Net } Benefit(p_t) = \frac{TP}{n} - \frac{FP}{n} \cdot \frac{p_t}{1 - p_t}, \tag{9}$$

where TP and FP denote the numbers of true positives and false positives, respectively, and n is the total number of individuals. We evaluate net benefit across a range of clinically relevant threshold probabilities. We compare HemoPain-Ensemble with two default reference strategies (treat-all and treat-none)[32] and with clinically motivated comparator models, including a single-biomarker reference model (ENS-RankPlatt using IL-10) and a six-biomarker blood model [30,32].

These procedures define the evaluation framework used in this study [28,29].

2.6 | Model interpretability

To improve transparency and clinical interpretability, we perform post hoc explainability analyses for the prespecified final model, HemoPain-Ensemble, only [33,34]. All interpretability analyses are conducted after model development and evaluation. These analyses are used to characterize model behavior, not to infer causality [33].

We quantify global feature importance using SHAP (Shapley Additive Explanations) values [34]. SHAP provides additive feature attributions at the individual level [34]. We aggregate SHAP values across samples to rank features by overall contribution. We also summarize effect direction to indicate whether higher or lower feature values are associated with higher predicted risk [34].

We further examine the structure of feature contributions in the SHAP attribution space.[35]. We calculate pairwise correlations between SHAP values across features. We then apply hierarchical clustering to the correlation matrix to identify groups of features with coordinated attribution patterns [36]. This analysis is used to explore coupled behavior among hematological features, beyond correlations in the original feature space.

We explore non-additive effects using SHAP interaction values for selected feature pairs [34,35]. This analysis assesses whether the contribution of one feature depends on the level of another feature.

We generate individual-level explanations to illustrate case-level reasoning [37]. Under the prespecified decision threshold, representative cases are examined using SHAP decompositions [34]. These examples show how multiple hematological features jointly shift predictions toward lower- or higher-risk categories.

Finally, we construct a surrogate decision tree in a strictly post hoc manner [33,36]. The

surrogate model is trained to approximate HemoPain-Ensemble predictions. It is used only as a communication tool to summarize key feature thresholds and dominant decision paths [37]. It does not contribute to prediction, model selection, or performance evaluation [33].

Together, these analyses provide a multi-level description of how hematological features contribute to model predictions [36].

3 | Results

3.1 | Data preprocessing and imputation results

Missing data were present across the hematological biomarkers and required appropriate handling prior to model development. Multiple imputation strategies were compared to evaluate their impact on variable distributions and inter-variable relationships. These comparisons were conducted within the development workflow to avoid information leakage.

As shown in Figure 2A, the distributions of the six raw hematological biomarkers after imputation differed across methods. Compared with mean imputation, MICE, and KNN, VAE-based imputation more closely preserved the shape and variability of the original observed distributions, without evidence of excessive variance shrinkage or artificial tail distortion.

Quantitative comparison using a composite imputation error metric demonstrated that VAE-based imputation achieved the lowest overall error across biomarkers (Figure 2B), indicating improved recovery of missing values relative to conventional methods [38]. Mean and MICE imputation showed intermediate performance, whereas KNN exhibited greater variability.

Preservation of biologically meaningful relationships among biomarkers was further assessed by examining correlation structures after imputation. As illustrated in Figure 2C, expected associations between inflammatory markers and HPA-axis–related hormones were maintained following VAE-based imputation, indicating that major inter-variable dependencies were preserved after imputation.

FIGURE 2 | Comparison of imputation methods for hematological biomarkers

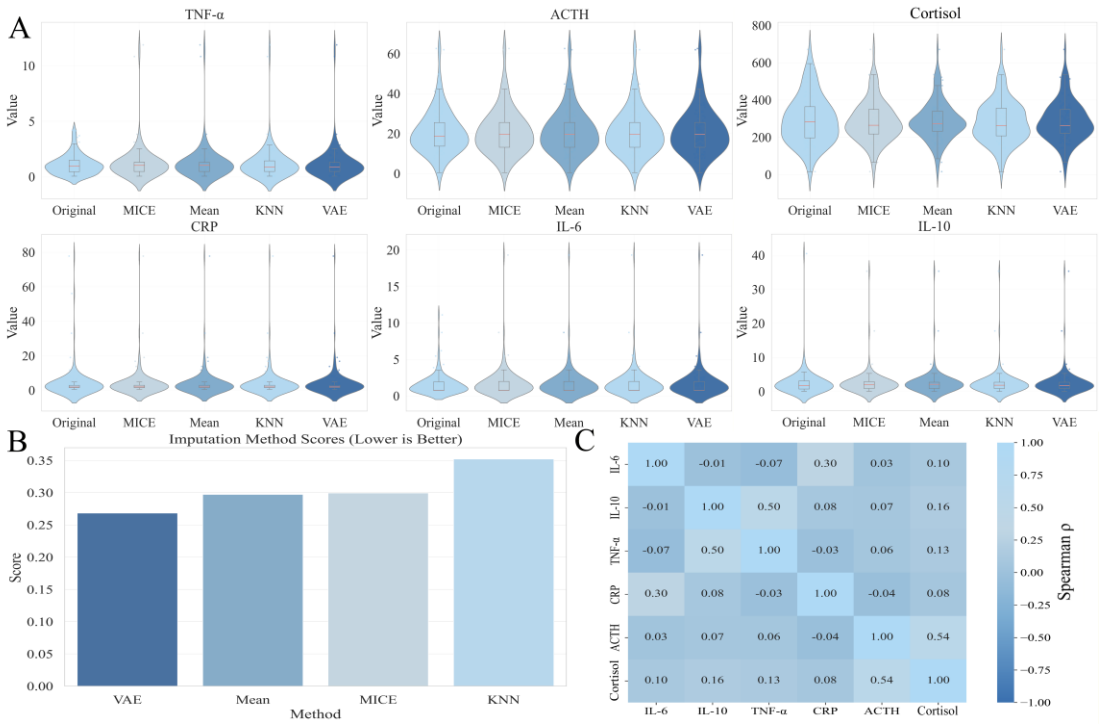

FIGURE 2 | Comparison of imputation methods for hematological biomarkers. (A) Violin plots showing the distributions of representative biomarkers (TNF-α, ACTH, Cortisol, CRP, IL-6, IL-10) in the original data and after imputation using MICE, mean imputation, KNN, and VAE. (B) Overall imputation performance scores for each method (lower is better). (C) Spearman correlation matrix among biomarkers after imputation.

Accordingly, VAE-based imputation is used in the downstream analyses reported below.

3.2 | Model comparison and selection

Figure 3A summarizes cross-validated model performance in the training set [39,40]. Gradient Boosting and Random Forest consistently ranked among the top individual learners across discrimination, calibration-related metrics, and stability during training-set development. Therefore, they were pre-specified as the base learners for subsequent ensemble construction [41]. Table 2 then reports out-of-sample performance of individual and ensemble models on the independent test set [42].

Although ENS-LogitT-AUC achieved the highest cross-validated AUC during training, ENS-LogitT was selected as the final model based on a joint assessment of discrimination, calibration-related metrics, and cross-fold stability, rather than on AUC alone [43].

Descriptively, Gradient Boosting achieved the highest test-set discrimination among individual models (AUC = 0.799; AP = 0.826), whereas Random Forest showed the highest sensitivity (Sen = 0.872) with a competitive F1 score (0.631).

TABLE 2 | Performance of machine learning models based on the 15-feature hematological panel on the

independent test set. Reported metrics include AUC, average precision (AP), Brier score, accuracy (Acc), sensitivity (Sen), specificity (Spe), F1-score, and negative predictive value (NPV). HemoPain-Ensemble denotes the final selected ensemble model.

Model	AUC	AP	Brier	Acc	Sen	Spe	F1	NPV
Logistic Regression	0.635	0.654	0.260	0.559	0.936	0.400	0.557	0.937
SVM	0.677	0.691	0.340	0.537	**0.950**	0.363	0.549	**0.945**
Neural Network	0.706	0.758	0.368	0.645	0.404	0.747	0.403	0.748
ENS-Stack	0.743	0.765	**0.217**	0.741	0.829	0.704	0.655	0.907
ENS-LogitT-AUC	0.773	0.791	0.351	0.720	0.702	0.728	0.598	0.853
XGBoost	0.775	0.792	0.349	0.720	0.702	0.728	0.598	0.853
LightGBM	0.776	0.808	0.360	0.677	0.446	**0.774**	0.450	0.768
CatBoost	0.779	0.807	0.337	**0.784**	0.680	0.828	0.651	0.860
Random Forest	0.782	0.785	0.265	0.698	0.872	0.625	0.631	0.920
ENS-RankPlatt	0.794	0.805	0.313	0.752	0.787	0.737	0.653	0.891
ENS-Top3	0.798	0.805	0.306	0.731	0.829	0.690	0.646	0.905
Gradient Boosting	0.799	0.826	0.350	0.698	0.489	**0.786**	0.490	0.785
HemoPain-Ensemble	**0.819**	**0.826**	0.295	0.763	0.787	0.753	**0.663**	0.893

Across all candidates, the logit-based convex ensemble with temperature scaling (HemoPain-Ensemble) demonstrates the best overall performance, yielding an AUC of 0.819, an AP of 0.826, and an F1 score of 0.663 [44].

To assess robustness, we apply 1,000 non-parametric bootstrap resamples to HemoPain-Ensemble on the test set (Figure 3B). The resulting distributions are narrow, and 95% confidence intervals show limited dispersion, indicating stable performance [42,45].

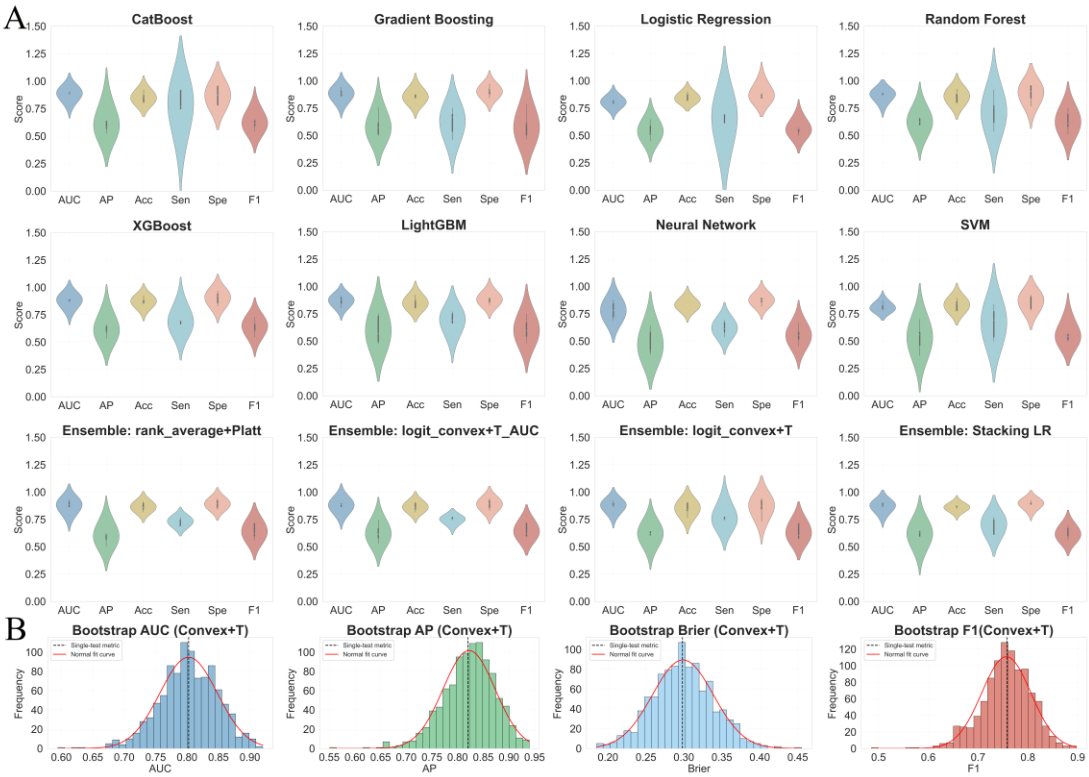

FIGURE 3 | Model comparison in the training set and bootstrap stability analysis of the final ensemble model. (A) Distributions of performance metrics (AUC, AP, Acc, Sen, Spe, and F1-score) across resamples for individual models and ensemble variants. (B) Bootstrap distributions of AUC, AP, Brier score, and F1-score for the final ensemble model.

Complete test-set performance of all models under the primary training-derived operating threshold is summarized in Supplementary Table S2; additional sensitivity analyses and bootstrap intervals are reported in Supplementary Tables S3–S5.

3.3 | Calibration and clinical utility

Calibration and clinical utility of the final model (HemoPain-Ensemble) were evaluated on the independent test set (Figure 4). The calibration plot suggested reasonable agreement between predicted probabilities and observed event rates across risk strata, without obvious systematic overestimation or underestimation. On the test set, the final model achieved an AUC of 0.819 and a Brier score of 0.295. Taken together, these results support good discriminative performance with acceptable probability estimation [46].

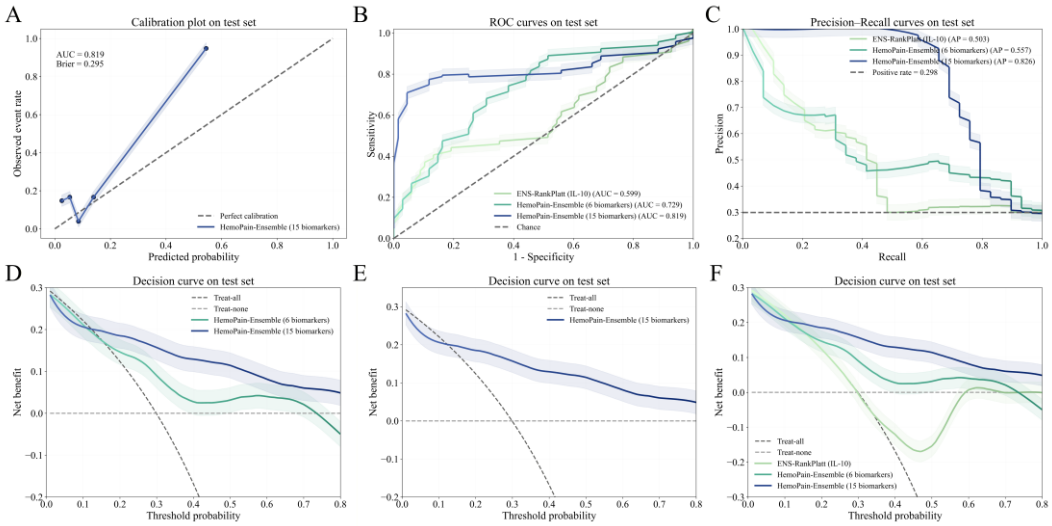

FIGURE 4 | Calibration, discrimination, and decision curve analysis on the independent test set (n = 97). (A) Calibration plot comparing observed event rates with mean predicted probabilities using K-bin grouping; the shaded area indicates bootstrap 95% confidence intervals (1,000 resamples).(B) ROC curves and (C) precision – recall curves for the final model and reference models.(D – F) Decision curve analysis (DCA) showing net benefit across threshold probabilities; shaded bands indicate bootstrap 95% confidence intervals (1,000 non-parametric resamples of the test set). Treat-all and treat-none strategies are shown as references. Interpretation of net benefit focuses on the mid-range thresholds where clinical decision-making is most relevant; uncertainty increases toward extreme thresholds due to the moderate test-set size.

Receiver operating characteristic and precision–recall curves further confirmed the robustness of the final model (Figure 4B–C). Compared with two clinically motivated reference models—ENS-RankPlatt (IL-10) and a six-biomarker blood model — HemoPain-Ensemble (15 biomarkers) demonstrated superior discrimination (AUC = 0.819 vs. 0.599 and 0.729). This advantage was also reflected in the precision–recall analysis (AP = 0.826 vs. 0.503 and 0.557; positive rate = 0.298), which is informative in settings with moderate class imbalance.

Clinical utility was assessed using decision curve analysis across a range of clinically relevant threshold probabilities (Figure 4D–F). The final model yielded a higher net benefit than treat-all and treat-none strategies across clinically relevant thresholds. Consistent with the ROC/PR results, HemoPain-Ensemble (15 biomarkers) showed higher net benefit than the six-biomarker model (Figure 4D) and ENS-RankPlatt (IL-10) (Figure 4F) for most clinically relevant threshold probabilities, although uncertainty increased at extreme thresholds due to the moderate test-set size (n = 97), as reflected by the confidence bands [47].

Favorable calibration and net benefit were achieved using the blood-based feature panel alone, without incorporating additional demographic or clinical scale variables into the primary model [48,49].

3.4 | Model interpretability and feature-level findings

To elucidate how the final model integrates the 15 hematological features for blood-based risk

stratification of current chronic pain status, a series of post hoc interpretability analyses were conducted for the prespecified final model.

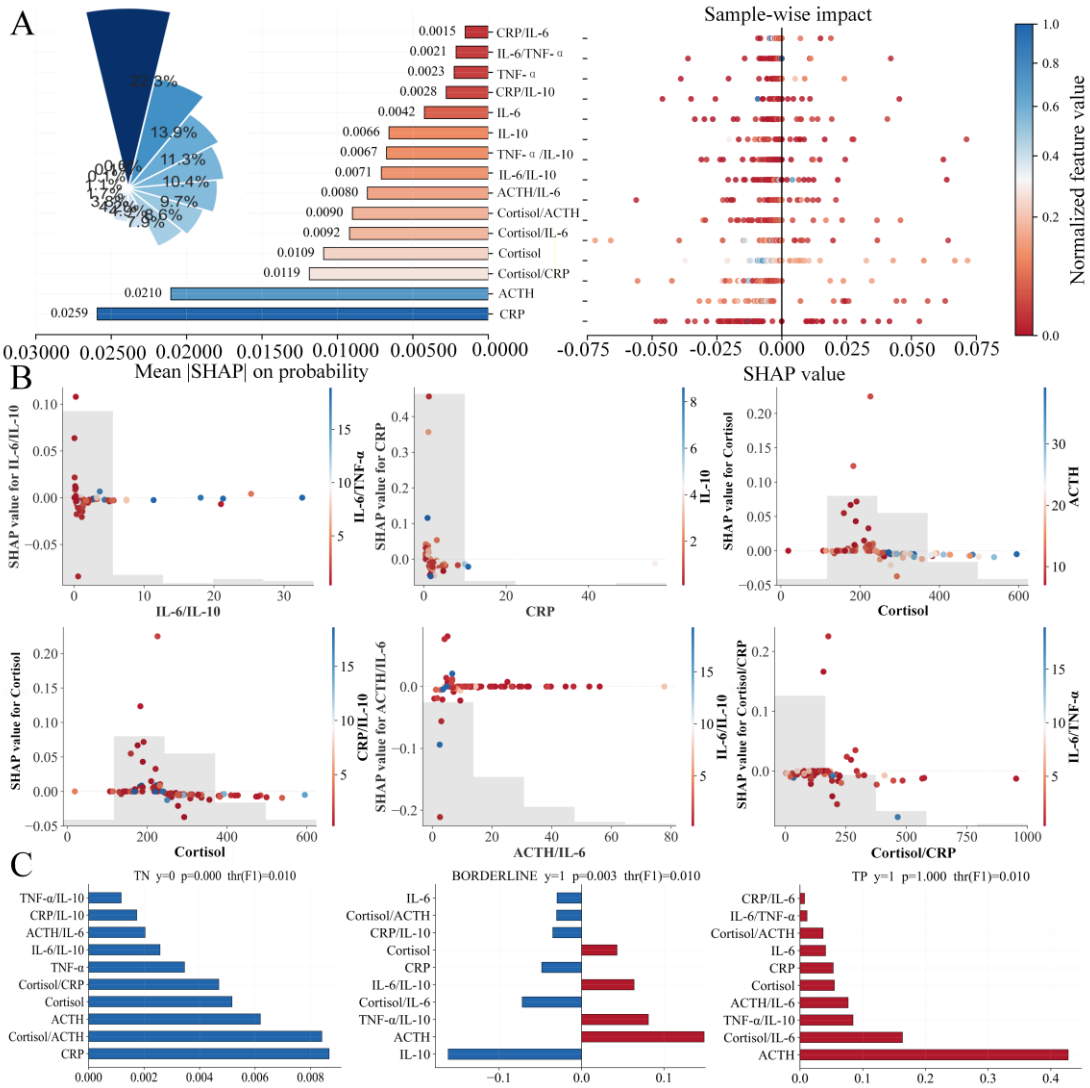

FIGURE 5 | SHAP-based interpretability analysis of the HemoPain-Ensemble model. (A) Global feature importance ranked by mean absolute SHAP values and sample-wise SHAP distributions. (B) SHAP dependence plots illustrating how key biomarkers/ratios influence predicted risk. (C) Group-wise SHAP comparisons across outcome strata, highlighting features with the largest directionally consistent contributions.

At the cohort level, SHAP-based attribution revealed a consistent hierarchy of feature contributions, with inflammatory markers and their derived ratios contributing most prominently to the predicted risk distribution (Figure 5A). Several ratio-based features, particularly those linking inflammatory activity to neuroendocrine regulation, showed larger absolute contributions than individual biomarkers alone. The directionality of SHAP values further demonstrated that elevations in pro-inflammatory dominance or altered HPA-axis balance were generally associated with increased predicted risk [50].

Beyond marginal importance, analysis of SHAP value correlations exposed structured

relationships among feature contributions (Figure 5B). Features clustered into coordinated attribution groups corresponding broadly to inflammatory processes and HPA-axis–related regulation [51–53]. These contribution-level associations were distinct from the correlation patterns observed in the raw feature space and were evident in the SHAP attribution structure.

To complement population-level patterns, individual-level explanations were examined under the fixed decision threshold [54]. SHAP decompositions for representative test-set samples illustrated how combinations of hematological features jointly drove predictions toward higher- or lower-risk classifications (Figure 5C). In high-risk cases, cumulative positive contributions were dominated by inflammatory ratios and cortisol-related features, whereas low-risk predictions reflected balanced or counteracting contributions across these domains. These profiles illustrate how feature contributions differ across individual predictions.

Finally, to provide a simplified representation of the ensemble's decision logic, a surrogate decision tree was used to approximate the final model's predictions (Figure 6). The extracted structure highlighted a limited number of recurrent split patterns centered on inflammatory ratios and plasma total cortisol–related thresholds. Leaf coverage analysis showed that these dominant paths encompassed a substantial proportion of test-set samples, suggesting the summarized decision rules capture common prediction behavior rather than isolated cases[55].

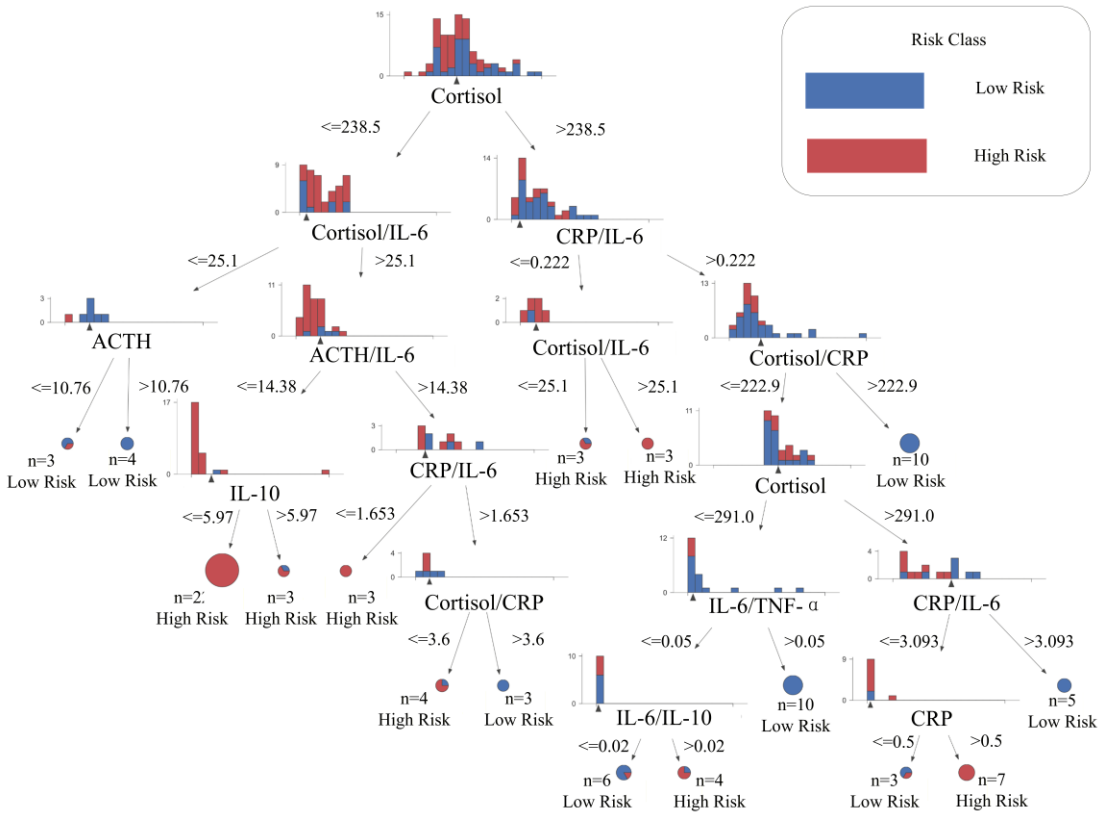

FIGURE 6 | Post hoc surrogate decision tree approximating ensemble decision logic. A shallow surrogate tree is fitted to the ensemble predictions to provide human-interpretable decision rules; node histograms show feature distributions, and leaf nodes indicate predicted risk classes (low vs. high).

Group-level attribution analyses, presented in the Supplementary Material, further showed that inflammatory and HPA-related feature groups accounted for most of the total contribution, with ratio-based features comprising a substantial proportion of explained risk (Supplementary Figure S2). Together, these analyses summarize the attribution patterns learned by the final model across inflammatory and HPA-axis–related features [52].

4. Discussion

在本研究中，我们首次报告了失眠障碍患者外周炎症标志物（IL-6、IL-10、TNF-α 和 CRP）、HPA-轴激素（皮质醇和 ACTH）与慢性疼痛之间的关系，以及各单项生物标志物对该人群慢性疼痛的诊断价值。为更好地识别这些患者的慢性疼痛，我们还利用这些生物标志物及生物标志物比值开发并验证了一种基于机器学习的多生物标志物组合模型。

慢性疼痛在失眠患者中非常普遍。已有研究证实，失眠与慢性疼痛之间存在潜在的双向联系，一旦形成共病，容易陷入相互加重的恶性循环，导致症状更复杂、治疗难度增加、预后更差。然而，准确、早期地检测失眠患者的慢性疼痛仍然具有挑战性。传统的慢性疼痛诊断方法主观性强且结果多变，缺乏客观依据，并且极易受到患者情绪和认知状态等因素的影响，这在失眠患者中尤其明显。但是，目前针对失眠患者中慢性疼痛的生物标志物的研究特别有限，因此，尽快开发和鉴定针对失眠患者慢性疼痛的客观生物标志物和诊断工具，对于早期准确诊断、制定有效的治疗策略以及评估预后和治疗效果至关重要。基于疾病发生的潜在机制来筛选生物标志物，被视作一种切实有效的策略，因为这些标志物具有高度可解释性并且可能具有潜在的高敏感性与特异性。此外，理想的诊断性生物标志物应兼具以下特性：样本易于获取、采集方式无创或微创、检测方法简便、成本低廉、结果可重复、患者依从性好及便于随访。既往研究提示炎症反应和 HPA 轴功能障碍可能是失眠患者共病慢性疼痛的重要潜在机制。因此，本研究选定以血液为检测样本，并聚焦于以下与失眠及慢性疼痛密切相关的生物标志物：IL-6, IL-10, TNF-α, CRP，ACTH, Cortisol。

本研究分析了来自四川大学华西医院的 481 例慢性失眠患者，其中 143 例（29.7%）伴有慢性疼痛。对比分析显示，相较于不伴慢性疼痛的失眠患者，合并慢性疼痛的患者整体健康状况更差、主观睡眠质量更低、失眠严重程度更高，且焦虑与抑郁水平也更显著。这一结果与既往研究相符，提示共病慢性疼痛的失眠患者临床表现更为复杂，进一步强调了在失眠群体中早期、准确识别并发慢性疼痛的临床重要性。仅对比六项血液指标发现，在两组患者比较中，仅 ACTH 在失眠伴慢性疼痛患者中统计学上显著小于失眠不伴慢性疼痛患者。回归分析显示仅 ACTH 与失眠患者慢性疼痛患病率存在相关性趋势。既往研究提示失眠患者存在促炎状态（如 IL-6, TNF-a 升高）并且 HPA 轴功能紊乱（皮质

醇、ACTH 水平失调）。促炎细胞因子如白细胞介素-6 (IL-6) 和肿瘤坏死因子α (TNF-α) 已被发现是强效的致痛剂和促痛剂，能够敏化初级感觉神经元，HPA 轴功能障碍会失去对炎症的调控并促进疼痛中枢敏化。动物研究发现 IL-6 与睡眠剥夺后疼痛敏感性呈正相关。在本项研究中，ACTH 具有显著变化，但是 PTC 并未出现明显变化，这可能提示 HPA 轴调节出现紊乱。而炎症指标似乎与既往的推测是矛盾的，但是这是可以理解的。首先，既往研究多是基于睡眠剥夺实验或对独立的失眠和慢性疼痛研究结果的联合推测，而尚未有研究直接对比失眠共病慢性疼痛与不伴疼痛患者的外周炎症和 HPA 轴指标。睡眠剥夺模型下的生理应激反应，与临床慢性失眠患者的长期适应状态存在本质差异。同样，将来自独立人群的结论简单叠加，可能无法准确反映两者共病时复杂的交互机制。其次，本项研究是一项横断面研究，外周单个指标的显著变化可能在失眠患者出现疼痛的早期。而在本研究的患者群体中，疼痛已慢性化（病程≥3 个月）。长期共病可能导致生理系统从初始的“高反应状态”（如普遍性的炎症指标升高）演变为一种更为复杂的“失调状态”。此外，慢性疼痛的维持可能更依赖于局部环境的变化，在外周循环中，单个外周血指标可能无法敏感捕捉到这些关键的局部变化。当然，这与样本量也密切相关，外周单个指标的变化可能较小，在样本量未达到理想状态下，可能导致统计效能不足，不能发现存在的潜在差异。但是，未发现外周血单个指标的普遍差异，并不等于“炎症机制和 HPA 轴紊乱”不参与。单个指标的绝对值有时很难去确定机体的病例生理状态，因为其在疾病发生和维持过程中可能起到的作用有限并且常无法捕捉标志物间的平衡和互相作用。标志物的比值可能更能反映炎症状态、HPA 轴稳态及炎症与 HPA 轴之间的互作。通过对比指标比值我们可以发现，失眠伴慢性疼痛患者 ACTH/IL6 比值显著低于失眠不伴慢性疼痛患者，PTC/ACTH 比值具有升高的趋势。而回归分析发现，TNFa/IL10 和 PTC/ACTH 比值增加，ACTH/IL6 比值降低与失眠患者慢性疼痛患病率增加有关。这些结果充分反映了失眠患者共病慢性疼痛是与炎症网络失调、HPA 轴紊乱及炎症/HPA 轴互作失调有关的。

在本项研究中，我们还重点关注了指标的诊断性能。我们选择了多种机器学习算法（Logistic Regression、SVM、Random Forest、XGBoost、Gradient Boosting、LightGBM、CatBoost 和 Neural Network 等）评估 6 项单个指标的诊断性能，发现单个指标的诊断价值均较低。虽然前述 ACTH 有统计学上的差异，但是其诊断效能并不高。此外，我们分析了不同指标组合对失眠患者共病慢性疼痛的诊断价值，因为多项指标的组合可能具有提升诊断效能的潜力。然而，本研究发现在小于 6 项指标构建的模型中，其诊断效能均不理想（AUC＜0.7）,当由 6 项指标共同诊断并集成 Gradient Boosting 和 Random Forest 算法时，其 AUC 可达到 0.729，敏感性 0.827，特异性 0.573。通过对诊断性能进行评价，该模型可达到 A2 级别，具有良好的诊断性能。然而，考虑到标志物之间存在平衡、通路互作等，我们创新性的将标志物比值纳入进了模型框架。这些标志物比值均已在既往文献中报道过，并且在其他疾病中具有一定的诊断价值。纳入这些新标志物后，集成 Gradient Boosting +Random Forest 算法构建的模型，其 AUC 达到 0.819，敏感性 0.787，特异性 0.753，整体诊断性能较 6 项指标明显提升，可以被评为 A1 级别，这反应了模型

极高的诊断性能。而这性能的显著提升充分反映了引入标志物比值在揭示复杂生理网络交互中的独特优势。与单一指标的线性信息不同，比值指标能够捕捉生物标志物之间的动态平衡与协同作用关系，更灵敏地反映出失眠共病慢性疼痛患者体内紊乱的神经内分泌与免疫调节通路。因此，我们的研究表明，在多指标联合诊断框架中，融入经过科学验证的比值标志物，能够更全面、更深入地解析疾病的多维病理生理特征，从而显著增强模型的判别能力与临床适用性。这一策略为未来开发高精度、可解释的疾病辅助诊断工具提供了重要的方法论参考。

进一步的可解释性分析提示失眠共病慢性疼痛的发生并非由炎症或 HPA 轴独立主导。模型通过整合炎症与神经内分泌信号之间的相互关系，形成个体化的风险判别模式。这一发现与当前将慢性疼痛视为免疫–内分泌系统相互作用结果的理论框架相一致，尤其在睡眠障碍背景下具有重要意义。（SHAP 分析需要进一步讨论）

具体而言，SHAP 分析进一步揭示了模型内部的判别逻辑及各类生物标志物对预测结果的相对贡献。从整体特征重要性来看，除部分单项指标外，多项比值特征在模型中的贡献更为突出，提示相较于单个炎症因子或单个 HPA 轴激素的绝对水平，反映炎症网络平衡、HPA 轴调节状态及二者相互作用的比值信息，更能够捕捉失眠患者共病慢性疼痛的关键生物学特征。这一点与我们前述结果相吻合：单个外周血指标的诊断价值有限，但将其置于更高层次的网络关系中后，其判别能力明显增强。进一步分析发现，特征贡献并非彼此孤立，而是呈现出一定的聚类和协同模式，即炎症相关指标及其比值、HPA 轴相关指标及其比值在模型中形成了相互联系的归因结构。这提示失眠共病慢性疼痛并不是由某一个单一通路异常所驱动，而更可能是促炎状态、抗炎调节不足、HPA 轴功能失衡以及炎症–神经内分泌互作紊乱共同作用的结果。与此同时，个体层面的 SHAP 分解还表明，不同患者达到高风险预测的路径并不完全相同：部分患者主要表现为炎症网络失衡驱动，另一些患者则更多体现为 HPA 轴相关异常及其与炎症信号的叠加作用。这说明失眠共病慢性疼痛在生物学层面可能具有一定异质性，也进一步提示基于多标志物组合而非单一指标进行风险识别更符合该疾病状态的真实病理生理特征。因此，SHAP 分析不仅增强了模型结果的可解释性，也从另一个角度支持了本研究的核心观点，即失眠患者共病慢性疼痛本质上是一种涉及免疫炎症调节、HPA 轴稳态及二者交互失衡的复杂生物学状态，而比值特征的引入恰恰有助于更敏感地表征这一复杂网络失调过程。

本研究具有以下优势。首先，本研究首次在失眠患者中证实了，相比于失眠不伴慢性疼痛患者，更高的炎症网络失调、HPA 轴紊乱（调节功能障碍）及炎症与 HPA 轴互作可能是失眠患者共病慢性疼痛的潜在机制。虽然单个指标中仅 ACTH 具有统计学上的显著差异，但是我们通过引入炎症指标比值，HPA 轴激素指标比值及炎症指标与 HPA 轴激素比值来反应机体的炎症状态，HPA 轴调节功能和炎症与 HPA 轴互作来证实了促炎状态，HPA 轴失调及炎症与 HPA 轴互作失调均参与了失眠共病慢性疼痛的发生。这为失眠共病慢性疼痛的发病机制提供了重要的研究证据。其次,本研究建立了首个基于大

样本、多指标、具有高度可解释性，融合机器学习算法，并通过测试验证的失眠共病慢性疼痛模型，具有优异的诊断性能，经评价体系可评估为 A1 级别。相较于依赖复杂和高成本技术（如基因测序、影像学检查等）获取的生物学指标，该体系所用指标易于采集与检测，成本低廉，患者接受度更高，可重复评估，更具备临床推广潜力。此外，在方法学层面，本研究展示了将临床动机明确的特征构建与受控集成学习相结合的优势。所有特征均基于生物学假设预先定义，避免了纯数据驱动的特征膨胀；集成模型仅基于性能最优的两个基础模型构建，并严格限制所有优化过程在训练集内完成，从而降低过拟合风险。结合独立测试集评估、bootstrap 稳定性分析及决策曲线分析，该框架强调模型的稳健性与临床相关性，而非单一指标的极致优化。

我们的研究确实存在一些局限性。首先，仅分析了炎症和 HPA 轴相关的指标，未对其他潜在机制的指标进行分析。其次，分析使用的是单中心数据（样本量有限），这可能限制我们研究结果的普适性。然而，通过采用广泛认可的标准操作程序、试剂和技术，我们预计在其他环境下也可以取得类似的结果。我们研究中观察到的稳健的 AUC 值支持了这一预期，表明疾病预测具有较高的准确性。这些结果验证了我们模型在临床诊断场景中的有效性，即使是在数据范围受限的情况下。为了确保更广泛的适用性和医学界的接受度，未来的研究必须在多中心环境中复制和验证我们的发现至关重要。此外，我们的研究并没有将临床特征、人口学及临床量数据纳入模型。因为本研究的首要创新目标之一是开发并验证一个基于客观体液生物标志物的诊断工具。我们假设，一个理想的'生物学标签'应能直接反映疾病的核心病理生理状态，而尽可能减少对社会人口学变量的依赖，以增强其在未知个体上的适用性。因此，我们决定优先保证模型的简洁性和生物学解释的纯粹性，构建一个完全基于生物标志物的诊断模型。

综上所述，我们的研究不但从失眠患者中验证了失眠患者共病慢性疼痛的核心发病机制，并且构建了具有良好诊断效能的诊断模型。在临床实践中，该模型有望发展为辅助决策的客观工具，帮助医生早期识别失眠患者中具有慢性疼痛高风险的个体，从而实现早期干预。在机制研究层面，该模型不仅是对 HPA 轴紊乱及炎症反应等理论假说的有力验证，也为深入揭示失眠与共病之间复杂的网络式生物学机制提供了新的研究线索与潜在靶点。

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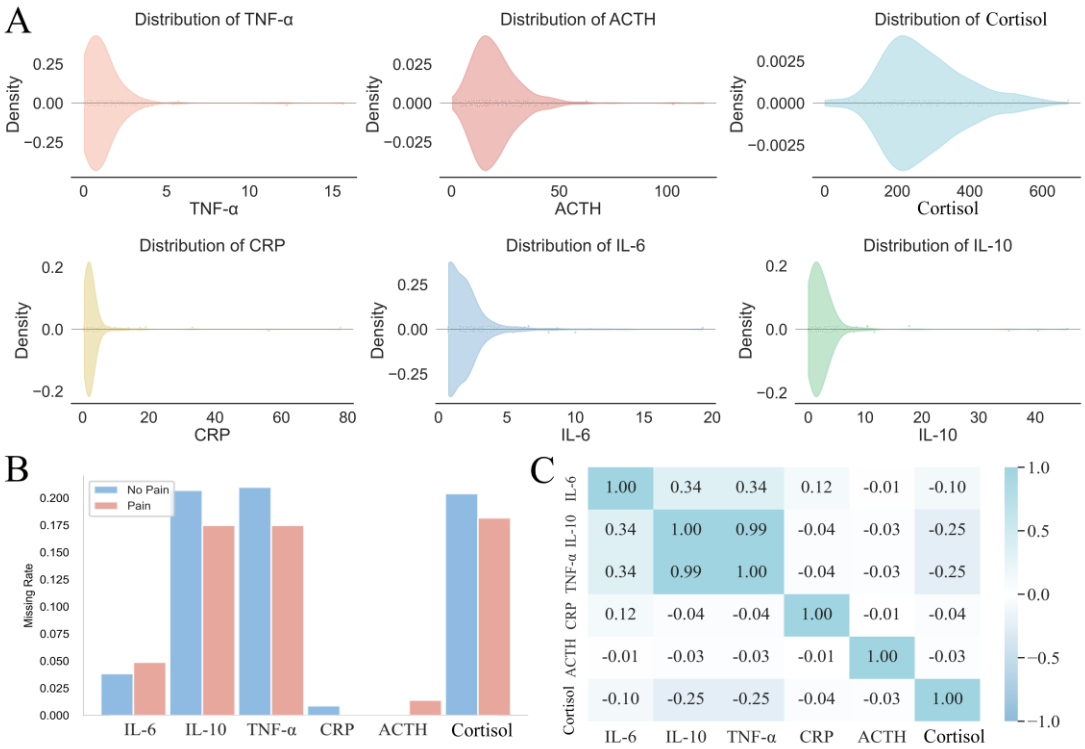
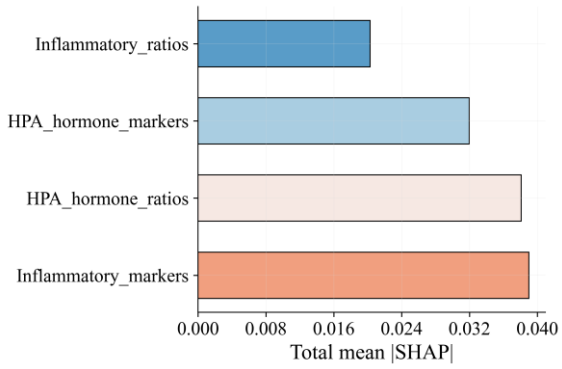
Supplementary Information



Supplementary Figure S1 | Pre-imputation distributions, missingness patterns, and correlation structure of the six primary blood biomarkers.(A) Empirical distributions of observed values for TNF-α, ACTH, cortisol, CRP, IL-6, and IL-10, shown as half-violin density plots with individual observations overlaid. (B) Missingness rate of each biomarker stratified by chronic pain status (Pain vs No Pain). (C) Pairwise Pearson correlation heatmap among the six biomarkers, computed from available pre-imputation data using pairwise complete observations; numbers denote correlation coefficients.



Supplementary Figure S2 | Relative SHAP contribution structure of inflammatory and HPA-axis–related feature groups. This figure summarizes the proportional attribution structure of inflammatory markers, inflammatory ratio features, HPA hormone markers, and HPA-related ratio features, quantified by the total mean absolute SHAP values within each feature group.

Supplementary TABLE S1 | Summary of demographic, lifestyle, and clinical variables and missingness. Missing values in blood biomarkers and derived ratio features were imputed using the mask-aware variational autoencoder (VAE) described in the Methods. The imputation model was fitted on the training set and applied unchanged to the test set. Because demographic and clinical scale variables were not incorporated into the primary model, their missingness is reported here for completeness.

Feature group	Variable	Missing n (%)
Demographic and lifestyle	Sex	1(0.2%)
Demographic and lifestyle	Age	1(0.2%)
Demographic and lifestyle	BMI	1(0.2%)
Demographic and lifestyle	Smoking_status	1(0.2%)
Demographic and lifestyle	Drinking_status	1(0.2%)
Demographic and lifestyle	Tea_drinking_status	1(0.2%)
Demographic and lifestyle	Coffee_drinking_status	1(0.2%)
Demographic and lifestyle	Education	1(0.2%)
Demographic and lifestyle	Occupation	1(0.2%)
Demographic and lifestyle	Marriage status	1(0.2%)
Demographic and lifestyle	Activity level	1(0.2%)
Demographic and lifestyle	ACEs_total_score_11	1(0.2%)
Clinical scales and status	Health_status_3groups	1(0.2%)
Clinical scales and status	Insomnia_duration	116(24.1%)
Clinical scales and status	ISI_total_score	1(0.2%)
Clinical scales and status	PSQI_total_score	1(0.2%)
Clinical scales and status	Epworth_total_score	1(0.2%)
Clinical scales and status	MEQ5_severity	1(0.2%)
Clinical scales and status	Chalder_14_total_score	1(0.2%)
Clinical scales and status	HAMD17_total_score	1(0.2%)
Clinical scales and status	HAMA_total_score	1(0.2%)
Clinical scales and status	Cognition screening	1(0.2%)

Supplementary TABLE S2 | Test-set performance of all models under the primary training-derived decision threshold. Performance metrics were computed on the independent test set using the primary decision threshold determined during model development on the training set.

Model	AUC	AP	Brier	Acc	Prec	Rec	Sen	Spe	F1	PPV	NPV	Youden
Logistic Regression	0.635	0.654	0.260	0.559	0.397	0.936	0.936	0.400	0.557	0.397	0.937	0.336
SVM	0.677	0.691	0.340	0.537	0.386	0.950	0.950	0.363	0.549	0.386	0.945	0.313
Neural Network	0.706	0.758	0.368	0.645	0.402	0.404	0.404	0.747	0.403	0.402	0.748	0.151
ENS-Stack	0.743	0.765	0.217	0.741	0.541	0.829	0.829	0.704	0.655	0.541	0.907	0.533
ENS-LogitT-AUC	0.773	0.791	0.351	0.720	0.521	0.702	0.702	0.728	0.598	0.521	0.853	0.430
XGBoost	0.775	0.792	0.349	0.720	0.521	0.702	0.702	0.728	0.598	0.521	0.853	0.430
LightGBM	0.776	0.808	0.360	0.677	0.455	0.446	0.446	0.774	0.450	0.455	0.768	0.220
CatBoost	0.779	0.807	0.337	0.784	0.625	0.680	0.68	0.828	0.651	0.625	0.860	0.508
Random Forest	0.782	0.785	0.265	0.698	0.495	0.872	0.872	0.625	0.631	0.495	0.920	0.497
ENS-RankPlatt	0.794	0.805	0.313	0.752	0.558	0.787	0.787	0.737	0.653	0.558	0.891	0.524
ENS-Top3	0.798	0.805	0.306	0.731	0.530	0.829	0.829	0.690	0.646	0.530	0.905	0.519
Gradient Boosting	0.799	0.826	0.350	0.698	0.491	0.489	0.489	0.786	0.490	0.491	0.785	0.275
HemoPain-Ensemble	0.819	0.826	0.295	0.763	0.573	0.787	0.787	0.753	0.663	0.573	0.893	0.54

Supplementary TABLE S3 | Cross-validation performance of all models in the training set. Values are reported as mean ± standard deviation across five cross-validation folds in the training set.

Model	AUC (mean±SD)	AP (mean±SD)	Brier (mean±SD)	Acc (mean±SD)	Prec (mean±SD)	Rec (mean±SD)	Sen (mean±SD)	Spe (mean±SD)	F1 (mean±SD)	PPV (mean±SD)	NPV (mean±SD)	Youden (mean±SD)
Neural Network	0.782±0.078	0.518±0.120	0.145±0.025	0.834±0.041	0.507±0.106	0.619±0.062	0.619±0.062	0.876±0.045	0.554±0.079	0.507±0.106	0.921±0.014	0.495±0.085
Logistic Regression	0.805±0.040	0.543±0.075	0.173±0.028	0.844±0.037	0.595±0.229	0.633±0.185	0.633±0.185	0.885±0.066	0.565±0.062	0.595±0.229	0.927±0.027	0.518±0.136
SVM	0.810±0.036	0.533±0.126	0.113±0.011	0.831±0.051	0.507±0.097	0.667±0.144	0.667±0.144	0.863±0.068	0.566±0.086	0.507±0.097	0.931±0.025	0.530±0.129
Random Forest	0.866±0.046	0.627±0.090	0.102±0.012	0.857±0.056	0.589±0.148	0.718±0.147	0.718±0.147	0.885±0.079	0.629±0.092	0.589±0.148	0.942±0.029	0.603±0.118
ENS-Stack	0.867±0.044	0.615±0.091	0.163±0.017	0.865±0.037	0.586±0.107	0.700±0.078	0.700±0.078	0.898±0.044	0.633±0.076	0.586±0.107	0.938±0.016	0.598±0.082
LightGBM	0.869±0.043	0.604±0.130	0.123±0.039	0.852±0.049	0.552±0.127	0.683±0.091	0.683±0.091	0.885±0.047	0.607±0.104	0.552±0.127	0.934±0.021	0.568±0.121
ENS-Top3	0.871±0.049	0.610±0.103	0.100±0.019	0.865±0.050	0.610±0.147	0.763±0.195	0.763±0.195	0.885±0.086	0.652±0.090	0.610±0.147	0.953±0.037	0.648±0.136
HemoPain-Ensemble	0.873±0.048	0.630±0.084	0.102±0.011	0.847±0.060	0.555±0.135	0.792±0.120	0.792±0.120	0.857±0.089	0.639±0.078	0.555±0.135	0.957±0.025	0.649±0.071
ENS-RankPlatt	0.874±0.055	0.607±0.086	0.099±0.010	0.855±0.056	0.568±0.114	0.792±0.120	0.792±0.120	0.866±0.088	0.649±0.063	0.568±0.114	0.958±0.025	0.659±0.058
XGBoost	0.876±0.052	0.628±0.095	0.105±0.020	0.870±0.043	0.616±0.129	0.696±0.095	0.696±0.095	0.903±0.058	0.643±0.069	0.616±0.129	0.939±0.016	0.600±0.079
Gradient Boosting	0.879±0.053	0.592±0.105	0.134±0.025	0.857±0.052	0.575±0.149	0.622±0.118	0.622±0.118	0.904±0.048	0.593±0.124	0.575±0.149	0.924±0.026	0.525±0.144
CatBoost	0.882±0.052	0.627±0.115	0.106±0.022	0.852±0.049	0.575±0.133	0.764±0.187	0.764±0.187	0.870±0.085	0.630±0.076	0.575±0.133	0.952±0.034	0.634±0.125
ENS-LogitT-AUC	0.888±0.056	0.629±0.089	0.113±0.010	0.855±0.056	0.568±0.114	0.792±0.120	0.792±0.120	0.866±0.088	0.649±0.063	0.568±0.114	0.958±0.025	0.659±0.058

Supplementary TABLE S4 | Sensitivity analysis under alternative decision threshold strategies. Model performance is reported under the primary training-derived threshold, a recall-constrained threshold (Recall≥80%), and the threshold maximizing the Youden index.

Model	Primary			Recall≥80%									Youden-max								
Model	AUC	AP	Brier	Acc	Prec	Rec	Sen	Spe	F1	PPV	NPV	Youden	Acc	Prec	Rec	Sen	Spe	F1	PPV	NPV	Youden
Logistic Regression	0.635	0.654	0.260	0.570	0.396	0.851	0.851	0.451	0.540	0.396	0.878	0.302	0.624	0.385	0.447	0.447	0.699	0.414	0.385	0.750	0.146
SVM	0.677	0.691	0.340	0.538	0.387	0.950	0.950	0.364	0.550	0.387	0.945	0.314	0.634	0.404	0.489	0.489	0.695	0.442	0.404	0.763	0.184
Neural Network	0.706	0.758	0.368	0.645	0.402	0.404	0.404	0.747	0.403	0.402	0.748	0.151	0.645	0.393	0.362	0.362	0.764	0.377	0.393	0.740	0.126
ENS-Stack	0.743	0.765	0.217	0.742	0.543	0.830	0.830	0.705	0.656	0.543	0.908	0.535	0.742	0.543	0.830	0.830	0.705	0.656	0.543	0.908	0.535
ENS-LogitT-AUC	0.773	0.791	0.351	0.720	0.521	0.702	0.702	0.728	0.598	0.521	0.853	0.430	0.720	0.521	0.702	0.702	0.728	0.598	0.521	0.853	0.430
XGBoost	0.775	0.792	0.349	0.720	0.521	0.702	0.702	0.728	0.598	0.521	0.853	0.430	0.720	0.521	0.702	0.702	0.728	0.598	0.521	0.853	0.430
LightGBM	0.776	0.808	0.360	0.677	0.455	0.447	0.447	0.774	0.451	0.455	0.768	0.221	0.677	0.455	0.447	0.447	0.774	0.451	0.455	0.768	0.221
CatBoost	0.779	0.807	0.337	0.763	0.581	0.723	0.723	0.780	0.644	0.581	0.870	0.503	0.785	0.627	0.681	0.681	0.829	0.653	0.627	0.860	0.510
Random Forest	0.782	0.785	0.265	0.720	0.518	0.809	0.809	0.682	0.632	0.518	0.894	0.491	0.731	0.532	0.766	0.766	0.716	0.628	0.532	0.879	0.482
ENS-RankPlatt	0.794	0.805	0.313	0.720	0.518	0.809	0.809	0.682	0.632	0.518	0.894	0.491	0.753	0.561	0.766	0.766	0.748	0.648	0.561	0.883	0.514
ENS-Top3	0.798	0.805	0.306	0.731	0.530	0.830	0.830	0.689	0.647	0.530	0.906	0.519	0.742	0.547	0.766	0.766	0.732	0.638	0.547	0.881	0.498
Gradient Boosting	0.799	0.826	0.350	0.699	0.493	0.489	0.489	0.788	0.491	0.493	0.785	0.277	0.699	0.493	0.489	0.489	0.788	0.491	0.493	0.785	0.277
HemoPain-Ensemble	0.819	0.826	0.295	0.699	0.496	0.830	0.830	0.644	0.621	0.496	0.900	0.474	0.763	0.573	0.787	0.787	0.753	0.663	0.573	0.893	0.540

Supplementary TABLE S5 | Test-set performance and bootstrap 95% confidence intervals for the final ensemble model. Bootstrap confidence intervals were estimated using 1,000 non-parametric resamples of the test set.

Metric	Test_point	Bootstrap_mean	Bootstrap_SD	CI_lower_2.5%	CI_upper_97.5%
AUC	0.819	0.819	0.045	0.710	0.889
AP	0.826	0.826	0.050	0.717	0.909
Brier	0.295	0.295	0.040	0.220	0.376
Acc	0.763	0.763	0.045	0.655	0.838
Sen	0.787	0.787	0.061	0.640	0.883
Spe	0.753	0.753	0.067	0.622	0.876
F1	0.663	0.663	0.049	0.544	0.737
PPV	0.573	0.573	0.063	0.435	0.693
NPV	0.893	0.893	0.064	0.743	0.993

Supplementary TABLE S 6 | Test-set performance of the best-performing model for each single blood biomarker or derived ratio feature. For each feature, multiple candidate models were trained during model development. To keep the table concise, we report on the independent test set only the single candidate model that achieved the highest test-set AUC for that feature.

Feature	Model	AUC	AP	Brier	Acc	Prec	Rec	Sen	Spe	F1	PPV	NPV	Youden
ACTH	Gradient Boosting	0.595	0.407	0.240	0.515	0.339	0.655	0.655	0.455	0.447	0.339	0.756	0.111
ACTH/IL-6	Light GBM	0.597	0.427	0.297	0.422	0.309	0.758	0.758	0.279	0.439	0.309	0.730	0.038
CRP	Logistic Regression	0.552	0.354	0.250	0.371	0.313	0.931	0.931	0.132	0.469	0.313	0.818	0.063
CRP/IL-10	DNN	0.575	0.385	0.228	0.298	0.298	1.000	1.000	0.000	0.460	0.298	0.000	0.000
CRP/IL-6	CatBoost	0.594	0.519	0.207	0.402	0.276	0.620	0.620	0.308	0.382	0.276	0.656	-0.070
IL-10	**ENS-RankPlatt**	**0.599**	**0.503**	**0.241**	**0.474**	**0.310**	**0.620**	**0.620**	**0.411**	**0.413**	**0.310**	**0.717**	**0.032**
IL-6	XGBoost	0.594	0.386	0.226	0.298	0.298	1.000	1.000	0.000	0.460	0.298	0.000	0.000
IL-6/IL-10	CatBoost	0.584	0.426	0.229	0.309	0.302	1.000	1.000	0.014	0.464	0.302	1.000	0.014
IL-6/TNF-α	Light GBM	0.594	0.402	0.252	0.298	0.298	1.000	1.000	0.000	0.460	0.298	0.000	0.000
cortisol	ENS-Stack	0.576	0.416	0.241	0.649	0.391	0.310	0.310	0.794	0.346	0.391	0.729	0.104
cortisol/ACTH	SVM	0.570	0.357	0.233	0.298	0.298	1.000	1.000	0.000	0.460	0.298	0.000	0.000
cortisol/CRP	DNN	0.599	0.357	0.226	0.505	0.362	0.862	0.862	0.352	0.510	0.362	0.857	0.215
cortisol/IL-6	HemoPain-Ensemble	0.594	0.435	0.202	0.587	0.342	0.413	0.413	0.661	0.375	0.342	0.725	0.075
TNF-α	Random Forest	0.562	0.372	0.233	0.350	0.302	0.896	0.896	0.117	0.452	0.302	0.727	0.014
TNF-α /IL-10	Gradient Boosting	0.593	0.566	0.221	0.484	0.352	0.862	0.862	0.323	0.500	0.352	0.846	0.185

Supplementary TABLE S7 | Test-set performance of models based on the six blood biomarkers only. All models were developed using the six original blood biomarkers only, without demographic variables or biomarker-ratio features. Results are reported on the independent test set. All remaining metrics are threshold-dependent and were calculated at the pre-specified primary operating threshold unless otherwise stated.

Model	AUC	AP	Brier	Acc	Prec	Rec	Sen	Spe	F1	PPV	NPV	Youden
Neural Network	0.627	0.626	0.214	0.793	0.714	0.517	0.517	0.911	0.600	0.714	0.815	0.429
Random Forest	0.629	0.530	0.180	0.618	0.431	0.862	0.862	0.514	0.574	0.431	0.897	0.376
CatBoost	0.656	0.596	0.207	0.793	0.666	0.620	0.620	0.867	0.642	0.666	0.842	0.488
SVM	0.661	0.459	0.221	0.546	0.384	0.862	0.862	0.411	0.531	0.384	0.875	0.273
LightGBM	0.662	0.637	0.258	0.762	0.750	0.310	0.310	0.950	0.439	0.750	0.764	0.260
ENS-Top3	0.665	0.596	0.198	0.783	0.681	0.517	0.517	0.897	0.588	0.681	0.813	0.414
ENS-LogitT-AUC	0.666	0.616	0.258	0.731	0.538	0.724	0.724	0.735	0.617	0.538	0.862	0.459
ENS-Stack	0.673	0.614	0.200	0.711	0.511	0.758	0.758	0.691	0.611	0.511	0.870	0.449
Logistic Regression	0.694	0.475	0.223	0.505	0.373	0.965	0.965	0.308	0.538	0.373	0.954	0.274
Gradient Boosting	0.696	0.553	0.213	0.525	0.373	0.862	0.862	0.382	0.520	0.373	0.866	0.244
XGBoost	0.713	0.568	0.240	0.721	0.535	0.517	0.517	0.808	0.526	0.535	0.797	0.326
ENS-RankPlatt	0.726	0.544	0.211	0.690	0.488	0.758	0.758	0.661	0.594	0.488	0.865	0.420
HemoPain-Ensemble	0.729	0.557	0.185	0.649	0.452	0.827	0.827	0.573	0.585	0.452	0.886	0.401